Lesson notes · for you

Goals and limits: unit check

Optimization · Level 2 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Choices, limits and a goal
- Draw the plans that fit
- Whole numbers only
- Tell the computer

Activity: Activity: Model swap

1. Recall (3 min): Write one idea from each of the four lessons, in your own words.
2. Write (7 min): Write a model for a choice you make (snacks, homework time, a team): variables, at least two limits and a goal, plus two plans.
3. Swap (7 min): Swap with a partner. Check their plans against every limit, and add one rule their model is missing, such as a fairness rule.
4. Share (3 min): Share the best rule someone added to your model, and why it matters.

Discussion

- Which idea from this unit surprised you most?
- When is the cheapest or biggest plan not the best one?
- How would you explain a model to a younger student?

Check yourself

- One model I could write this week, and its goal.

Try it this week

Pick one choice you make every week and write it as a model: variables, limits and a goal. Check two plans, and add one rule that makes it fair to someone else.

Worksheet

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Name _____ Class _____ Date _____

1 **In a model, what is a variable?**

- ◯ A choice you decide, with a unit
- ◯ A rule every plan must keep
- ◯ The number you make as big as possible

2 **The garden: x tomato beds, y bean beds. Match each rule to its maths.**

Land: at most 8 beds	• • • •	30x + 40y
Water: at most 14 cans a day	• • • •	2x + y ≤ 14
The profit to make as big as possible	• • • •	x + y ≤ 8

3 **The plan 6 tomato beds and 2 bean beds keeps all three garden limits (land 8, water 14, hours 18).**

- ◯ True
- ◯ False

4 **In a linear program, where is the best plan?**

- ◯ At a corner of the shaded shape
- ◯ In the middle of the shaded shape
- ◯ Just outside the shaded shape

5 **Which choices can only be whole numbers or yes-or-no? Choose all that apply.**

- ☐ How many buses to book
- ☐ Whether the desks go on the truck
- ☐ Litres of juice for the party
- ☐ How many tomato beds to plant